AHUJA CAREER INSTITUTE

ACADEMIC YEAR 2026–2027
Since 1998 • Ahmedabad

11th & 12th Science Comprehensive Program

Physics • Chemistry • Mathematics • Biology (GSEB / CBSE Boards + JEE / NEET)

Target Board & Exams	Std. 11th & 12th Board Examinations & Competitive Entrance (JEE Main / NEET UG)	**Instruction Medium**	English & Gujarati Medium (Offline, Online & Hybrid)
Faculty Directorate	Senior Science Directorate & Entrance Specialists	**Class Timings**	Afternoon & Evening Batches: 3:00 PM – 9:00 PM (Special Morning Batches Available)
Assessment Frequency	Chapter-wise Tests, Fortnightly Board Drills, and Full-Syllabus Mock Exam Series	**Campuses Available**	Maninagar Head Office & Vastral Campus

PROGRAM CORE PEDAGOGY & DELIVERABLES

Key Feature: 100/100 Perfect Scores in Mathematics & Chemistry

Key Feature: NCERT first-principles conceptual derivations and extensive numerical drill sets

Key Feature: Daily Practice Problems (DPPs) with daily tracking and doubt resolution

Key Feature: Continuous 1-on-1 doubt solving desks available at Maninagar and Vastral campuses

SUBJECT: PHYSICS

Curriculum Scope: Mechanics, Heat, Electromagnetism, Optics, Semiconductors and Modern Physics.

No.	Chapter / Unit Title	Core Focus Topics & Exam Weightage Elements
1	**Units, Dimensions & Motion in a Straight Line**	SI Dimensional Analysis, Kinematics Equations, Calculus in Motion, Motion Graphs
2	**Laws of Motion, Work, Energy & Power**	Newton's 3 Laws, Friction Dynamics, Work-Energy Theorem, Elastic & Inelastic Collisions
3	**System of Particles & Rotational Motion**	Center of Mass, Torque, Moment of Inertia Theorems, Angular Momentum Conservation
4	**Thermodynamics & Kinetic Theory**	Thermal Expansion, Heat Engines, Carnot Cycle, Gas Laws & Mean Free Path
5	**Electrostatics, Magnetism & Wave Optics**	Gauss Law, Biot-Savart Law, Electromagnetic Induction, AC Circuits, Wave Interference

SUBJECT: CHEMISTRY

Curriculum Scope: Organic Reaction Mechanisms, Inorganic Periodic Properties, and Physical Chemistry Numericals.

No.	Chapter / Unit Title	Core Focus Topics & Exam Weightage Elements
1	**Basic Concepts of Chemistry & Atomic Structure**	Mole Concept, Stoichiometry, Bohr Model, Quantum Numbers & Electron Configurations
2	**Chemical Bonding & Molecular Structure**	VSEPR Theory, Hybridization, Molecular Orbital Theory, Dipole Moments
3	**Thermodynamics & Chemical Equilibrium**	Enthalpy, Entropy, Gibbs Free Energy, Le Chatelier Principle, pH & Buffer Solutions
4	**Organic Chemistry & Hydrocarbons**	IUPAC Nomenclature, Reaction Intermediates, Electrophilic Addition, Elimination Mechanisms
5	**Solutions, Electrochemistry & Coordination**	Raoult's Law, Colligative Properties, Nernst Equation, Crystal Field Theory

SUBJECT: MATHEMATICS

Curriculum Scope: Calculus, Vectors, Coordinate Geometry, Matrices, and Advanced Probability.

No.	Chapter / Unit Title	Core Focus Topics & Exam Weightage Elements
1	**Sets, Relations, Functions & Trigonometry**	Domain-Range Analysis, Injective/Surjective Mapping, Trigonometric Identities & Equations
2	**Complex Numbers, Matrices & Determinants**	Argand Plane, De Moivre's Theorem, Matrix Inverses, System of Equations (Cramer's Rule)
3	**Differential Calculus & Derivatives**	Limits, Continuity, Differentiability, Chain Rule, Maxima & Minima Applications
4	**Integral Calculus & Differential Equations**	Indefinite Integrals, Definite Integral Properties, Area Under Curves, Linear ODEs
5	**Coordinate Geometry, Vectors & 3D Lines/Planes**	Conic Sections, Vector Scalar/Cross Products, Shortest Distance Between Skew Lines

SUBJECT: BIOLOGY

Curriculum Scope: Human Physiology, Genetics, Cytology, Plant Physiology, and Ecology.

No.	Chapter / Unit Title	Core Focus Topics & Exam Weightage Elements
1	**Biological Classification & Diversity**	Five Kingdoms, Plant & Animal Kingdoms, Viral and Bacterial Structures
2	**Cell Biology & Cell Division**	Cell Organelles, Fluid Mosaic Model, Mitosis, Meiosis, Cell Cycle Regulation
3	**Human Physiology & Organ Systems**	Digestion, Respiration, Cardiac Cycle, Excretion, Nervous Impulse Transmission
4	**Genetics & Molecular Biology**	Mendelian Inheritance, DNA Replication, Transcription, Translation, Lac Operon
5	**Biotechnology & Ecology**	Recombinant DNA Technology, PCR, Ecosystem Energy Flow, Biodiversity Conservation

ADMISSIONS & INQUIRY HELPLINE:
• **Maninagar Head Office:** 4th Floor, Takshashila Square, Krishnabaug Cross Road, Maninagar — **+91 74053 28676**
• **Vastral Branch:** 502, Avadh Pride, Opp. Metro Pillar 140, Nirant Cross Road, Vastral — **+91 97243 19900**
Office Timings: Monday to Saturday, 10:00 AM – 6:00 PM • Website: ahujainstitute.com